

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

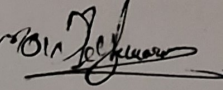
**Duration : 1 Hour
Total Marks:30**

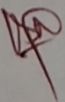
Annexure A

Constraint-Based Problem Solving

Code of Conduct:

D. MOUL ESHWARAN (12DS13077) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

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1. Map coloring problem (Backtracking search)
Given:

i) Colour = {Red, Green, Blue}

ii) Adjacent: $R_1 - R_2, R_2 - R_3, R_3 - R_4, R_4 - R_1$

iii) $R_1 = \text{Red}$

iv) $R_3 \neq \text{Blue}$

step by step

Step 1: Assign $R_1 = \text{Red}$

Step 2: Assign $R_2 = \text{Green}$ (cannot be Red because it is adjacent to R_1)

Step 3: Assign $R_3 = \text{Red}$

i) Red is allowed (R_2 is Green)

ii) Blue is not allowed

iii) so $R_3 = \text{Red}$

Step 4: Assign $R_4 = \text{Green}$

i) cannot be Red (adjacent to R_1)

ii) Green is allowed (R_3 is Red)

Final Coloring

R_1	Red
R_2	Green
R_3	Red
R_4	Green

Final Answer:

$R_1 = \text{Red}, R_2 = \text{Green}, R_3 = \text{Red}, R_4 = \text{Green}$

2. Form Investigation Robot (BFS search)

since the 6x6 grid is not given, only the general answer can be written.

steps:

- i) start froms.
- ii) expand all neighbouring nodes (up, down, left, right).
- iii) ignore obstacle cells.
- iv) continue until G is reached.
- v) BFS always finds the shortest path.

Example path

$S \rightarrow (1,2) \rightarrow (1,3) \rightarrow (2,3) \rightarrow (3,3) \rightarrow (4,3) \rightarrow (5,3)$
 $\rightarrow G$

Cost

i) Cost of each move = 1

ii) total moves = 7

total Cost = 7

Final Answer:

i) Algorithm: BFS

ii) optimal path: $S \rightarrow (1,2) \rightarrow (1,3) \rightarrow (2,3) \rightarrow (3,3) \rightarrow (4,3) \rightarrow (5,3) \rightarrow G$

iii) total Cost: 7

: If you see a question paper, provides a specific 6x6 grid, upload that page. The exact BFS.

- ii. Tech2 must be assigned before Tech3 (Machine X < Machine Y < Machine Z)
 - iii. Only one technician per machine
 - iv. Tech3 cannot repair Machine X
 - v. All technicians must be assigned exactly one machine
- Apply Backtracking Search to find a valid assignment, showing all intermediate steps and constraint checks. State the final valid schedule.

2. A delivery robot operates in a grid environment (5x5). It must move from Start (S) to Goal (G). Obstacles block certain cells.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using the above constraints and BFS Search Algorithm, show step-by-step node expansion and frontier updates, and determine the optimal path and cost.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for	Provides justification with some clarity	Weak or unclear justification	No justification provided

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